AYUSH CLINICAL INTELLIGENCE

Hypertension (Uccha Raktachap / Vyana Bala Vaishamya)

ICD-10: I10 | Evidence Grade: A–C per WHO Traditional Medicine Guidelines

Sources: CCRAS / AYUSH Research Portal / PMC / J-AIM Elsevier / JAIMS / Frontiers Pharmacology 2025

1. TRIAGE GATE — MANDATORY FIRST CHECK

This gate must be evaluated before any AYUSH intervention is surfaced. Conditions below require immediate referral to allopathic emergency care. AYUSH management protocols do NOT apply.

⚠ REFER IMMEDIATELY — DO NOT INITIATE AYUSH MANAGEMENT: • SBP > 180 mmHg with chest pain, visual changes, confusion, or proteinuria • Hypertension onset under age 20 • Sudden onset hypertension (any age) • Hypertension with suspected MI, acute angina, or stroke signs Source: CCRAS Cross-Referral Guidelines — ayushportal.nic.in

✓ AYUSH MANAGEMENT SCOPE: Essential (primary) hypertension — Stage 1: SBP 140–159 / DBP 90–99 mmHg Essential (primary) hypertension — Stage 2: SBP 160–179 / DBP 100–109 mmHg Age range with clinical evidence: 30–60 years | No active end-organ damage

2. DISEASE ENTITY MAPPING

FIELD	VALUE
Biomedical Term	Essential Hypertension
ICD-10 Code	I10
Ayurveda Term	Uccha Raktachap | Vyana Bala Vaishamya | Raktadhatugata Vata
Unani Term	Irtifa-e-Dababul Dam | Sumbul-ut-Teeb (drug correlate)
Siddha Term	Elevated Vatham | Innatham
Dosha Classification	Tridoshaja Vyadhi — primary vitiation: Vyana Vayu (cardiovascular Vata) with Pitta in Rakta Dhatu
Pathogenesis Stage	Prasara-Avastha: spread of vitiated Vyana Vata, Prana Vata, Sadhaka Pitta, Avalambaka Kapha with Rakta Dhatu

2.1 Clinical Symptom Entities for NLP Extraction

Prevalence data from JAIMS clinical study — Virechana Karma in Uccha Raktachapa, October 2024, n=30 EHT patients.

AYURVEDA TERM	BIOMEDICAL EQUIVALENT	PREVALENCE IN EHT
Shirahshoola	Headache	76.36%
Bhrama	Giddiness / Dizziness	65.45%
Anidra	Insomnia / Sleep disturbance	58.18%
Klama	Fatigue / Malaise	41.81%
Swedadhikya	Excess sweating	30.90%

3. PHARMACOLOGICAL INTERVENTIONS

3.1 M-Sarpagandha Mishran [Evidence Grade: A/B]

ATTRIBUTE	DETAIL
Type	Polyherbal compound — classical Ayurveda formulation
Ingredients	Sarpagandha root | Jatamamsi rhizome | Vacha rhizome | Punarnava root | Brahmi plant | Shankhpushpi plant | Guduchi stem
Adjunct — CCRAS Protocol	Praval Pishti 250mg (Purified Coral + Rose Water distillate) — vascular elasticity support via natural calcium
Dose — CCRAS NPCDCS	M-Sarpagandha Mishran 250mg + Praval Pishti 250mg — twice daily after meals with lukewarm water — 6 months
Dose — AIIMS RCT Protocol	500mg capsules twice daily — 12 weeks — add-on to Hydrochlorothiazide 12.5mg/day
Manufacturer	IMPCL (Indian Medicines Pharmaceutical Corporation Ltd) — GMP-certified, Government of India enterprise
Indication	Stage 1 essential hypertension — SBP 140–159 / DBP 90–99 mmHg | Age 18–70
Active Alkaloid	Reserpine (from Rauwolfia) — central + peripheral catecholamine depletion

Clinical Outcome Data — M-Sarpagandha Mishran

STUDY	DESIGN / n	OUTCOME	SIGNIFICANCE
Pilot OPT — AIIA New Delhi	Open label, n=30, 8 weeks	SBP: 158.53 → 126.6 mmHg DBP: 98.37 → 78.37 mmHg MBP: 118.43 → 84.43 mmHg	$p < 0.0001$ (paired t-test)
CCRAS NPCDCS Multicentric (Rajasthan, Bihar, Gujarat)	Retrospective observational n = multiple cohorts 6 months	Integrative group (Ayurveda + Yoga): 3 months: 44.92% dose reduction/discontinuation 6 months: 66.99% dose	Inter-group comparison statistically significant in Group II sub-group A

		reduction/discontinuation Lifestyle + Yoga only: 3 months: 27.78% 6 months: 48.14%	
AIIMS RCT (Active — CTRI/2021/12/038589)	Phase III double-blind double-dummy vs Amlodipine 5mg n=150 (75 per arm) 12 weeks	Primary: SBP/DBP change at week 12 Secondary: IL-6, Pro-BNP, oxidative stress, lipid profile, SF-36	Results pending — highest evidence RCT for AYUSH antihypertensive

⚠ **ADR FLAGS — M-SARPAGANDHA MISHRAN (Reserpine content): • Depression: 30/195 patients on Rauwolfia products reported depression in monitored cohort; 24/30 were female • Screen for depression history BEFORE prescribing • Contraindicated in pregnancy (uterine stimulant effect) • Do NOT combine with MAO inhibitors or tricyclic antidepressants • Monitor for bradycardia, sedation, nasal congestion**

3.2 Jatamamsi — Nardostachys jatamansi [Evidence Grade: A]

ATTRIBUTE	DETAIL
Cross-System Use	Ayurveda (Jatamamsi) + Unani (Sumbul-ut-Teeb) — referenced by Avicenna in Kitab al-Adwiya al-Qalbiya for cardiac diseases
Active Compounds	Jatamansone, nardostachone, valeranal — sesquiterpenes, alkaloids, coumarins, flavonoids
Mechanism	Vasodilatory + antioxidant + anti-inflammatory + anxiolytic + sedative + antispasmodic
Best Indication	Stress-driven hypertension with anxiety and insomnia as primary co-presenting symptoms
Dose — RCT	3g/day (1 capsule x 3 times daily) for 4 weeks
Study Design	Single-blind RCT, placebo-controlled, n=40, age 35–70, Stage 1 EHT on ≤2 antihypertensives
Outcome	SBP: 144.20 ± 11.35 → 134.30 ± 10.08 mmHg | DBP: significantly reduced
Significance	$p < 0.001$ | QoL measured via MINICHAL score
Source	J Ayurveda Integr Med — ScienceDirect / PubMed PMID: 33066862

3.3 Terminalia arjuna (Arjuna) [Evidence Grade: B]

ATTRIBUTE	DETAIL
Cross-System Use	Ayurveda + Siddha + Unani — bark decoction documented in Charaka Samhita, Sushruta Samhita, Ashtanga Hridayam
Active Compounds	Arjunolic acid, oligomeric proanthocyanidins (OPCs), arjunone, luteolin, gallic acid, ellagic acid, phytosterols

Mechanism	Antioxidant + anti-ischemic + vasodilatory + hypolipidemic — different from Sarpagandha (no catecholamine depletion)
Best Indication	BP elevation with concurrent dyslipidemia, cardiac remodeling, left ventricular dysfunction
Key Outcome — Abana Study	Significant SBP reduction + improved ejection fraction + reduced echocardiographic LV diameter, posterior wall thickness, IVS thickness
Formulations	Bark powder 500mg twice daily | Arjunarishta (fermented): 15–30ml with equal water after meals

⚠ **ADR FLAGS — ARJUNA: • Additive hypotension with conventional antihypertensives — monitor BP closely • Warfarin interaction — potential anticoagulant potentiation (bleeding risk) • Rare GI upset at high bark powder doses • Insufficient safety data in pregnancy — avoid**

3.4 Gokshura — Tribulus terrestris [Evidence Grade: B]

ATTRIBUTE	DETAIL
Mechanism	Diuretic — renal vasodilatory | Reduces fluid retention by increasing renal blood flow
Best Indication	Kapha-predominant hypertension with edema/fluid retention
Dose	3g/day Ghanasatwa (solid water extract) in three divided doses
Duration	4 weeks
Study Design	Clinical study, n=75, mild–moderate EHT, whole plant and fruit extracts compared
Outcome	Significant reduction in clinical symptoms, SBP, and DBP — both plant and fruit preparations effective — no side effects reported
Source	Ancient Science of Life — Murthy AR, Dubey SD, Tripathi K (2000)

3.5 Punarnava — Boerhavia diffusa [Evidence Grade: B]

ATTRIBUTE	DETAIL
Active Compounds	Liridodendrin, hypoxanthine, boeravinones
Mechanism	Calcium channel antagonist (via boeravinones) + diuretic via increased renal blood flow and smooth muscle relaxation
Best Indication	Hypertension with renal component, fluid overload, or hepatic involvement
Dose	250mg extract — 2 capsules twice daily with water
Duration	6 weeks

Outcome	SBP: 151.48 ± 5.75 → 137.33 ± 5.23 mmHg DBP: 95.41 ± 2.06 → 87.11 ± 4.75 mmHg
Source	Nayak et al., Ayushdhara 2(6):390–396, 2015

4. COMPOSITE POLYHERBAL FORMULATIONS

Source: Frontiers in Pharmacology meta-analysis, November 2025 (44 studies, 10 RCTs, n=524 participants) — doi: 10.3389/fphar.2025.1695614

FORMULATION	EVIDENCE BASE	NOTES
Hibiscus sabdariffa (Ambasthaki) tea/extract	Multiple RCTs — substantial BP reduction across independent trials	Widely replicable — tea preparation; dose varies by trial
Emblica officinalis (Amalaki) + conventional	RCT — significantly outperformed placebo as add-on	Ghaffari et al. 2020; antioxidant synergy with conventional drugs
Allium sativum / Rason (Garlic preparations)	Multiple arms — dose-dependent antihypertensive effect confirmed	Ashraf et al. 2013; Sobenin et al. 2009
Arjuna Vachadi Yoga (polyherbal)	Comparative study — better outcomes than CPS combination	vs Chandraprabhavati + Punarnava mandura + Sweta parpati
Karsha Vati	Statistically significant BP improvement	Classical formulation; limited modern replication
Balsamodendron mukul (Guggulu)	Significant improvement; dyslipidemia co-benefit	Thyroid interaction — screen before use
Boerhavia diffusa (Punarnava) extract	Significant BP improvement in standalone trials	See Section 3.5 for dose details
Baktadushtihar yoga + conventional antihypertensives	Combination improved outcomes over standalone therapy	Avhad et al. 2016 — synergistic approach

5. PANCHAKARMA PROCEDURES

5.1 Virechana Karma (Therapeutic Purgation) [Evidence Grade: B]

ATTRIBUTE	DETAIL
Mechanism	Eliminates vitiated Pitta from Rakta Dhatu via Anulomana action; corrects Srotosanga (channel obstruction) and Agnimandhya (digestive fire impairment)
Dosha Indication	Pitta-dominant EHT: heat symptoms, constipation, anger/irritability, redness — Raktadushti pattern
Post-Virechana Oral Medication	Arjuna Churna + Gokshura Churna (equal parts) — 10g twice daily with lukewarm water — 15 days

Study Design	Clinical study, n=30 EHT — JAIMS Oct 2024 (Patel et al.)
Evidence Principle	Shodhana + Shamana combination demonstrated better clinical response than Shamana (palliative) alone
Comparative Data	Virechana + Arjunadi ghana vati > Basti or Sarpagandha vati alone for SBP/DBP reduction — Frontiers meta-analysis 2025
Contraindication	Debilitated patients, severe dehydration, IBD, acute illness, pregnancy

5.2 Shirodhara (Continuous Oil Stream — Forehead) [Evidence Grade: B symptoms / C for BP]

ATTRIBUTE	DETAIL
Oils Used	Bala Taila (primary) | Brahmi Taila | Chandana Bala Taila | Takra (buttermilk = Takradhara variant)
Mechanism	Parasympathetic activation — calms Prana Vayu; reduces cortisol-driven vascular tension
Best Indication	Psychogenic / stress-driven hypertension; Manasika component prominent; insomnia, anxiety, giddiness
Meta-analysis — BP effect	Systematic review + meta-analysis 2024 — 4 RCTs, n=100: SBP: MD -1.257 mmHg, I^2 = 89.3% (not conclusive) DBP: MD -0.40 mmHg, I^2 = 0 (statistically significant but low clinical value)
Symptom Relief	Significantly higher in Shirodhara + Ayurveda drug vs Ayurveda drug alone — headache, insomnia, giddiness, palpitation, fatigue
Clinical Use	ADJUNCT to pharmacological intervention — not standalone BP management
Source	PMC 11918277 — Systematic Review and Meta-Analysis 2024

6. YOGA / PRANAYAMA — CCRAS CLINICAL PROTOCOL

Government-backed protocol with multicentric observational outcome data — NPCDCS-AYUSH Integration Project, 3 states, 2015–2019.

CCRAS NPCDCS Outcome — Yoga + Lifestyle alone (n = cohort B1): 3 months: 27.78% of patients achieved conventional drug dose reduction or discontinuation 6 months: 48.14% — reflecting cumulative benefit of sustained yoga practice

COMPONENT	DETAIL
Asana	Surya Namaskar | Seated asanas for cardiovascular stability
Kriya	Documented in CCRAS NPCDCS Table 1 — specific to cohort protocol

Pranayama — Primary	Nadi Shodhana (alternate nostril breathing) — parasympathetic activation, reduces sympathetic tone
Pranayama — Secondary	Bhramari (humming bee breath) — vagal stimulation, rapid stress reduction
Relaxation	Shavasana — structured corpse pose for full autonomic reset
Duration / Frequency	Daily practice — minimum 6 months for statistically significant BP reduction outcome
Integration	Combine with Ayurveda pharmacological intervention for maximum effect (66.99% vs 48.14% drug reduction at 6 months)

7. PHARMACOVIGILANCE — ADR SAFETY SIGNALS

Source: ayushsuraksha.com — National Pharmacovigilance Program for ASUH Drugs, Ministry of AYUSH.

DRUG / HERB	KNOWN ADRs	INTERACTION FLAGS
Sarpagandha / Rauwolfia	Depression (dose-related, higher risk female) Bradycardia Sedation Nasal congestion	MAO inhibitors — severe Tricyclics — severe Other antihypertensives — additive hypotension
Arjuna	Additive hypotension Rare GI upset	Warfarin — potentiation (bleeding risk) Conventional antihypertensives — monitor BP
Ashwagandha	Additive thyroid effect GI disturbance Sedation	Thyroid medications — use with caution CNS depressants — additive sedation Autoimmune conditions — avoid
Guggulu (Commiphora mukul)	GI irritation Allergic dermatitis (rare)	Thyroid hormones — interaction Anticoagulants — monitor
Punarnava	Generally well tolerated	Diuretics — additive effect — monitor electrolytes
Gokshura	Rare GI upset at high dose	Diuretics — additive; monitor renal function in compromised kidneys

8. ENTITY-TO-INTERVENTION ROUTING LOGIC

For use in clinical decision engine. Match extracted NLP entities to first-line AYUSH intervention and evidence grade.

EXTRACTED ENTITY CLUSTER	FIRST-LINE AYUSH INTERVENTION	EVIDENCE GRADE

Anxiety + Anidra + mild BP elevation	Jatamamsi 3g/day OR Sarpagandha Mishran (low dose) — screen for depression first	A
Edema + fluid retention + BP elevation	Gokshura Ghanasatwa 3g/day | Punarnava 250mg x 2 bd	B
Dyslipidemia + BP elevation + cardiac symptoms	Arjuna bark 500mg bd | Arjunarishta 15–30ml bd	B
Stress + headache + giddiness + insomnia	Shirodhara (Bala Taila) — adjunct + Brahmi vati	B (symptoms) / C (BP)
Pitta constitution + heat + constipation + BP elevation	Virechana Karma + Arjuna/Gokshura Churna post-procedure	B
Stage 1 EHT — wants conventional drug reduction	M-Sarpagandha Mishran 250mg + Praval Pishti 250mg bd + CCRAS Yoga protocol	A/B
Stage 1 EHT — integrative alongside allopathic	M-Sarpagandha Mishran 500mg bd (AIIMS dose) + Hydrochlorothiazide 12.5mg — CTRI protocol	A — trial ongoing
Kapha constitution + obesity + BP elevation	Guggulu-based formulation + lifestyle | Gokshura for diuresis	B

9. EVIDENCE GRADE SUMMARY

Evidence grading per WHO Traditional Medicine Research Guidelines (A = RCT, B = clinical studies without RCT, C = expert opinion / case reports).

INTERVENTION	EVIDENCE GRADE	SOURCE	N
M-Sarpagandha Mishran (AIIMS RCT — active)	**A (trial ongoing)**	CTRI/2021/12/038589 — PMC 11871589	150
M-Sarpagandha Mishran (CCRAS NPCDCS)	**B (retrospective observational)**	PMC 8377180 — J-AIM 2021	multicentric
Jatamamsi — Nardostachys jatamansi	**A (single-blind RCT)**	PMC/ScienceDirect — PMID 33066862	40
Terminalia arjuna (Arjuna)	**B (multiple clinical studies)**	PMC 4220499	multiple
Gokshura — Tribulus terrestris	**B (75-patient clinical study)**	Anc Sci Life 2000	75
Punarnava — Boerhavia diffusa	**B (6-week clinical study)**	Ayushdhara 2015	~50
Shirodhara — symptom relief	**B**	PMC 11918277 — Meta-analysis 2024	100
Shirodhara — BP reduction	**C (low clinical value)**	PMC 11918277	100
Virechana Karma + Shamana	**B**	JAIMS Oct 2024	30
Yoga + Pranayama (CCRAS protocol)	**B (multicentric observational)**	PMC 8377180	cohort

10. MASTER PROMPT — CLINICAL INTELLIGENCE EXTRACTION

Copy the prompt block below verbatim into the clinical engine or use as a system-level extraction directive for subsequent AYUSH disease modules.

```
// MASTER PROMPT — AYUSH CLINICAL INTELLIGENCE EXTRACTOR
// Version 1.0 | Disease Module: Hypertension (ICD-10 I10)

You are a clinical intelligence extraction engine for AYUSH (Ayurveda, Yoga,
Unani, Siddha, Homeopathy) medical systems. Your task is to extract and
structure verifiable clinical intelligence from the designated AYUSH sources
for a given disease entity. Output must be factual, source-attributed, and

evidence-graded. No vague descriptions. No wellness blog content.

DISEASE INPUT: {DISEASE_NAME} | ICD-10: {ICD_CODE}

AYUSH SYSTEMS: Ayurveda | Unani | Siddha | Yoga

SEARCH SOURCES IN ORDER:
1. ayushportal.nic.in          — clinical research, ICD-10 disease mapping, evidence grades
A/B/C
2. accr.ayush.gov.in           — expert-reviewed clinical case repository
3. dharaonline.org             — Ayurveda literature database
4. echarak.ayush.gov.in        — classical texts, botanical knowledge
5. ayushsuraksha.com           — pharmacovigilance, ADR signals
6. ccras.nic.in                — CCRAS official clinical research protocols
7. pubmed.ncbi.nlm.nih.gov     — filter by: Ayurveda OR Unani OR Siddha + disease
8. jaims.in                    — JAIMS peer-reviewed journal
9. sciencedirect.com (J-AIM)   — Elsevier Journal of Ayurveda and Integrative Medicine
10. frontiersin.org/pharmacology — systematic reviews / meta-analyses

EXTRACT THE FOLLOWING ENTITIES — NO VAGUE TEXT:

[1] TRIAGE GATE
    - Emergency referral conditions for this disease — list as binary flags
    - Scope boundary: which patient profile qualifies for AYUSH management


[2] DISEASE ENTITY MAP
    - Biomedical term + ICD-10 code
    - Ayurveda term + Unani term + Siddha term
    - Dosha classification + pathogenesis stage (Kriyakala)
    - NLP symptom entities: AYUSH term | biomedical equivalent | prevalence % if available


[3] PHARMACOLOGICAL INTERVENTIONS (per drug/herb)
    - Drug name (AYUSH + Latin binomial)
    - Cross-system use (Ayurveda / Unani / Siddha)
    - Active compounds (named)
    - Mechanism of action (specific — not generic)
    - Best-fit indication (dosha pattern or biomedical co-condition)
    - Exact dose used in clinical study
    - Duration of treatment
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    - Outcome data: baseline BP or parameter → post-treatment value ± SD, p-value
    - Study design (RCT / observational / pilot) and n
    - Source citation (PMC ID / DOI / journal name / government project name)
    - Evidence grade: A / B / C (WHO criteria)
    - ADR flags: named adverse effects + contraindications + drug interactions

[4] COMPOSITE FORMULATIONS
    - Formulation name + constituent herbs
    - Evidence base + outcome summary
    - Source

[5] PANCHAKARMA / PROCEDURAL THERAPIES
    - Procedure name + oils/materials used
    - Mechanism
    - Dosha indication
    - Quantified outcome data if available (MD, mmHg reduction)
    - Clinical use context: standalone vs adjunct
    - Contraindications
    - Evidence grade

[6] YOGA / PRANAYAMA PROTOCOL
    - Named techniques (not generic 'yoga')
    - Duration + frequency
    - Quantified outcome if available
    - Source / project name

[7] PHARMACOVIGILANCE (per drug)
    - Named ADRs
    - Drug interactions (named — not generic 'may interact')
    - Contraindicated conditions
    - Source: ayushsuraksha.com or published ADR data

[8] ENTITY-TO-INTERVENTION ROUTING TABLE
    - Column 1: Extracted NLP entity cluster
    - Column 2: First-line AYUSH intervention + dose
    - Column 3: Evidence grade

[9] EVIDENCE GRADE SUMMARY TABLE
    - All interventions listed with: Grade | Study design | n | Source citation

OUTPUT FORMAT RULES:
- Every claim must have a source URL, PMC ID, DOI, or government project reference
- State exact doses — never 'as directed' or 'standard dose'
- State exact BP changes — never 'significantly reduced'
- State evidence grade on every intervention — never omit
- Flag every ADR — never suppress for readability
- If evidence does not exist for a sub-entity, state: NO EVIDENCE FOUND
- Do not include wellness site content, commercial Ayurveda clinic claims,
  or content without traceable source to government or peer-reviewed database
- Do NOT surface any AYUSH intervention if triage gate conditions are met
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END OF DOCUMENT — AYUSH Clinical Intelligence | Hypertension Module | Generated: February 2026